

Z – Parameters

Aim:- To determine the Z- parameters of a two port resistive network.

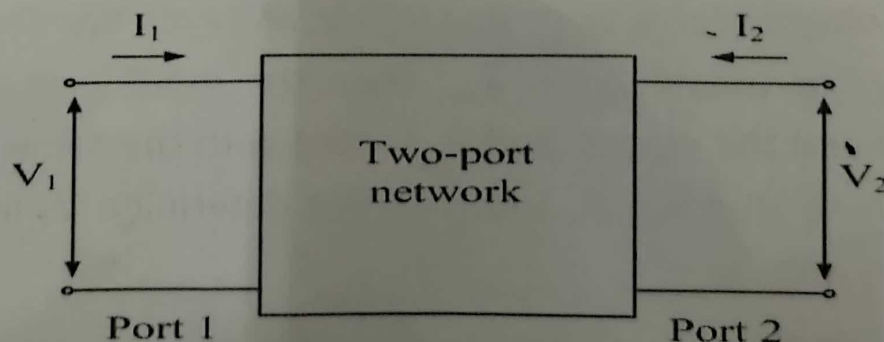
Apparatus Required:

- 1) Resistance 2) Digital Multimeter
- 3) Bread Board 4) Power Supply
- 5) Connecting wires

Theory:

A two-port network is a special case of multiport network. Each port consists of two terminals. One for entry and the other for exit. From the definition of a port the current at entry is equal to that at the exit terminal of a port.

In the two port networks there are four variables. These are the voltages and currents at the input and the output ports. Here only two of the four variables V_1 , I_1 , V_2 , I_2 are independent.



$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} \quad Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} \quad Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

Where,

Z_{11} – Input driving point impedance

Z_{12} – Reverse transfer impedance

Z_{21} – Forward transfer impedance

Z_{22} – output driving point impedance

Procedure:

In order to determine the open circuit (impedance-Z) parameters open the output port and excite input port with a known voltage source V_s so that $V_1 = V_s$ and $I_2=0$. We determine I_1 and V_2 to obtain Z_{11} & Z_{21} . Then the input port is open circuited and the output port is excited with the same voltage source V_s so that $V_2 = V_s$ and $I_1=0$. We determine V_1 and I_2 to obtain Z_{12} and Z_{22} .

Observation Table:

CASE – I When output port is open circuited, $I_2=0$

V_1 (v)	I_1 (A/mA)		V_2 (V)		$Z_{11}=V_1/I_1$		$Z_{21}=V_2/I_1$	
	Exp	Theo	Exp	Theo	Exp	Theo	Exp	Theo
Applied								

CASE – II When input port is open circuited, $I_1=0$

V_1 (v)	V_2 (V)		I_2 (A/mA)		$Z_{12}=V_1/I_2$		$Z_{22}=V_2/I_2$	
	Exp	Theo	Exp	Theo	Exp	Theo	Exp	Theo
Applied								

Result: Z-Parameters of a given network are as follows:

EXPERIMENTAL	THEORETICAL
$Z_{11} =$ $Z_{12} =$ $Z_{21} =$ $Z_{22} =$	